

HEALTH INFORMATION SYSTEM

OBJECTIVES

- Define HIS
- Data sources
- Types of HIS
- Data collection methods
- data storage and analysis
- information utilisation

DEFINITION

- Health information systems refers to any system that captures, stores, manages or transmits information related to the health of individuals or the activities of organisations that work within the health sector.
- Health information system is that that system in which collection, utilization, analysis and transmission of information is done for conducting health services, training and research.

CONT'

- It is essential for health system policy development and implementation, governance and regulation, health research, human resources development, health education and training, service delivery and financing.

KEY FUNCTIONS OF HIS

- I. Data generation
- II. Compilation
- III. Analysis and synthesis
- IV. Communication and use.

CONT

- The health information system collects data from health and other relevant sectors, analyses the data and ensures their overall quality, relevance and timeliness and converts the data into information for health-related decision-making.
- The key users of health information policy-makers, planners, managers, health-care providers, communities and individuals.

The broad objectives of HIS

1. Provide data for monitoring and evaluation;
2. Provide an alert and early warning capability;
3. Support patient and health facility management;
4. Enable planning;
5. Provide a basis for research
6. Facilitate health situation and trends analyses;
7. Facilitate reporting, and reinforce communication of health challenges to diverse users.

CONT'

Data: Is raw facts/real-world observations, events, transactions and so on which have been recorded

- Facts obtained by reading, observation, counting, measuring, weighing, etc which are then recorded.

Information: Is processed data for use

- Is knowledge and understanding that is usable by the recipient

Health information: is information about people's health status, resources, activities and events as related to health

System: A collection of components that work together in a regular or orderly manner, to achieve a common objective

Data Processing

- The goal of data processing is to present information in a way that aids/helps in decision making in all levels
- It is a process of transforming raw data to a state which analysis can be done smoothly with minimal distortion of results

Types of data

- Qualitative (Categorical data) either nominal or ordinal (Not measurable you can only describe, i.e. descriptive)
- Quantitative (Measurement/Numerical data) either discrete/interval or continuous (It is measurable)_

METHODS DATA COLLECTION

- I. Observation
- II. Questionnaires
- III. Interviews
- IV. Surveys
- V. Documents and records
- VI. Focus groups
- VII. Oral histories

METHODS OF DATA STORAGE

- A. On-Premises. Considered to be the original data storage method, an on-premises data solution typically involves servers that are owned and managed by the organization itself
- B. Colocation
- C. Cloud Storage
- D. Hard Disk Drives (HDD)
- E. Solid State Drive (SSD)
- F. Tape Drives
- G. 5D Data Storage.

Example of data

1.Activity data

- i. MCH/FP attendances
- ii. ANC attendant
- iii. IMM attendant
- iv. Postnatal attendant
- v. Outpatient attendant
- vi. Inpatient attendance

2.Epidemiological surveillance

- i. AFP surveillance
- ii. Measles surveillance
- iii. NNT surveillance

CONT'

3.Semi permanent data

- Pop target to be served
- Facility staffing
- Volunteers

4.Program data

- Stock cards
- Lab specimens
- No of CBD's

5.Other types of data that should be routinely collected

- Administrative data such as stock card for supplies ,lab specimens.
- Human resource e.g. number of staff, workload and absenteeism.
- Logistical data e.g. transport
- Organizational data e.g. infrastructure and equipment

The Ahima model (1998) includes following characteristics of quality data

- **Accessibility** - Data items should be easily obtainable. Should be **available** on time and at all levels (old data is of historical value only)- decisions must be made based on current information,.
- **Comprehensiveness**- It is complete. All required data items are included. Ensure the entire scope of data is collected
- **Accuracy** - Data are free from errors (typographical or misspelling)
- **Consistency** – Data need to be consistent in terms of collection, transmission and use of abbreviations
- **Representative** - It represents all recorders of similar data

CONT'

- **Comparable** - using the same definitions of items – if we don't measure by the same tool, we can't compare each other's results.
- **Timeliness**- available on time
- **Currency** – up to date Many types of healthcare data become obsolete after a period of time
- **Precision** – It denotes how close to an actual size, weight, or other standard a particular measurement. E.g. drug dosage must be very precise
- **Relevancy** – Collect data will be used

Types of HMIS

- A. Community Based
- B. Facility Based (e.g. In-pt HIS, Outpt HIS)
- C. District Based

Principle of HIS- Characteristics of well designed HIS

- Collecting less data – collects prioritized/essential data based on indicators on a routine basis
- Uses the data collected
- Non duplication of data collection
- Interdependence / Integration -various depts., units have to contribute or various player to cooperate
- Interoperability – ability of the system to communicate with other systems
- Availability of forms/reporting tools (Manual or modern e.g. computers, Smartphone's, PDA)
- Encourages decentralized planning and management

IMPORTANCE OF HIS

- A. Is essential for health system policy development and implementation, governance and regulation, health research, human resources development, health education and training, service delivery and financing.
- B. Gives a clear picture of health and sickness at all level of society – individual, household, community, county, country, regional and global.
- C. Helps prevent the spread of disease and improve the health of individuals.
- D. Leads to better decisions and better spending.
- E. Makes it easier to track and confront threats to health at all levels

DIFFERENT KINDS OF INFORMATION NEEDS

- The different kinds of information needed include:
 1. **Health determinants** -socioeconomic, environmental, behavioural and genetic factors) and the contextual environments within which the health system operates;
 2. **Inputs to the health system and related processes** - policy and organisation, health facilities and equipment, costs, human and financial resources and health information systems;
 3. **The performance or outputs of the health system**- availability, accessibility, quality and use of health information and services, responsiveness of the system to user needs, and financial risk protection;
 4. **Health outcomes** - mortality, morbidity, disease outbreaks, health status, disability and wellbeing
 5. **Health inequities**- determinants, coverage of use of services, and health outcomes, and including key stratifiers such as sex, socioeconomic status, ethnic group and geographical location.

DATA SOURCES

Type of Information Source

- I. Population based information** -Census, household surveys, specials studies, civil/vital registration systems
- II. Health service information-** Outline service statistics, national health accounts, electronic medical records service
- III. Health management information** -Financial information system, human resource information system, logistics and supplies information system, service statistics system, electronic medical records system, pharmaceutical procurement
- IV. Special health information** - Academic, research institutions and international organisations for research and survey findings

THREE COMPONENTS OF HIS

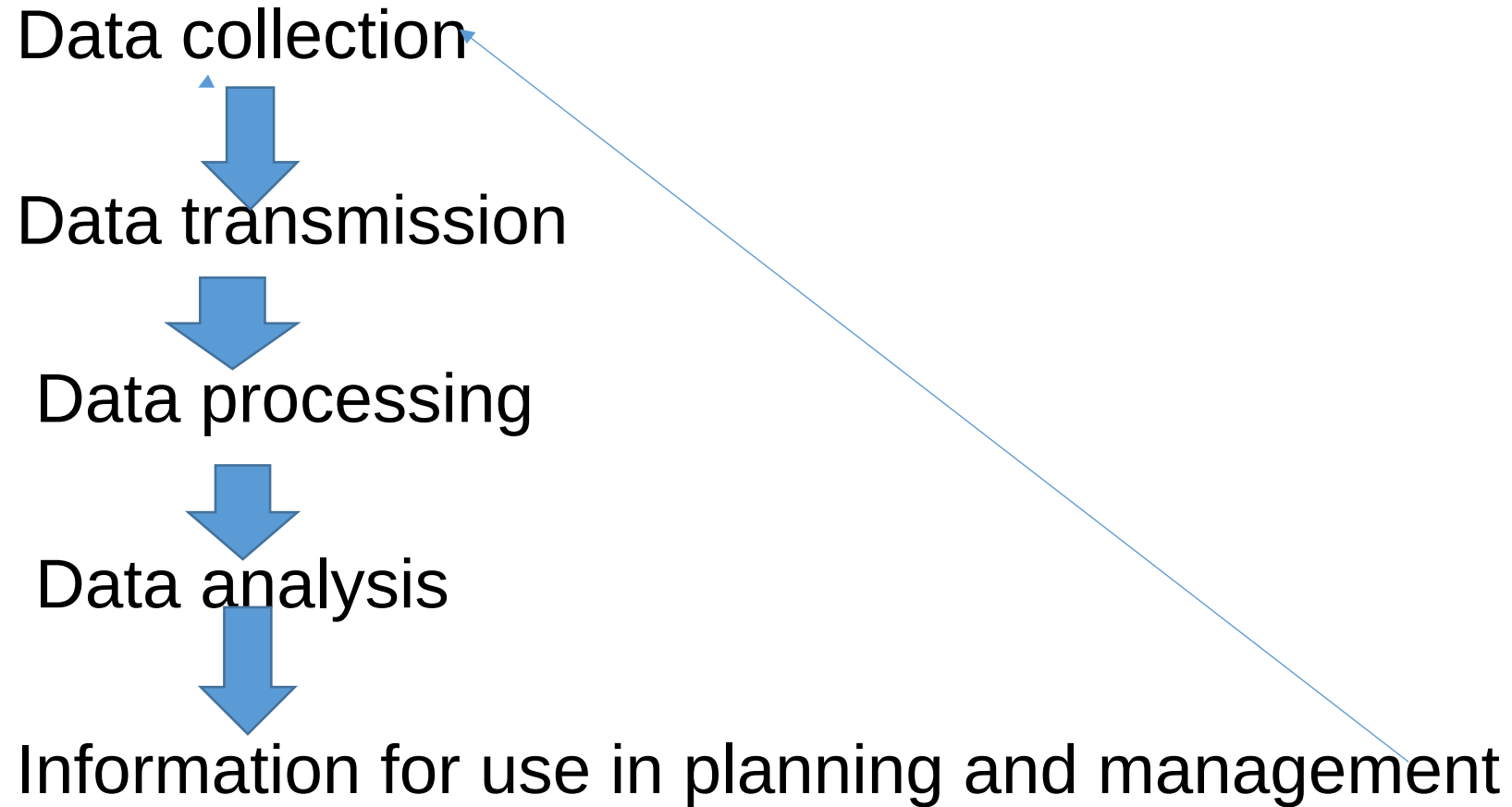
1. **INPUTS**- They include the ability of those responsible to lead and co-ordinate the process; the existence of necessary laws and policies; financial resources and people with the necessary skills to do the work and the infrastructure
2. **PROCESSES** – these includes
 - i. Indicators – a set of measures that show changes in health profile;
 - ii. Data sources – an integrated HIS brings together data from a variety of sources;
3. **OUTPUTS**- The information produced – the HIS outputs – should be relevant, accessible, and useful evidence for decision making.

Information products are collated from a range of sources, and synthesised into usable statistics that can be analysed and compared;

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Dissemination and use. Through widespread dissemination and use of information products, the HIS provides direct benefit to all those who participate in it, providing an ongoing incentive for users to continue to strengthen the system.

INFORMATION PROCESS



The Kenya National Health Information system (NHIS)

- is a comprehensive and integrated structure that collects, collates, analyses, evaluates, stores, disseminates health and health-related data and information for use by all.
- The NHIS seeks to address such issues as partnership in data collection and information sharing, guidelines on data processing and data warehousing as well as instituting standardised mandatory reporting by all care providers (public and private) and quality in data management in the health sector

Composition of NHIS

- producers of health statistics including Ministry of Health, Kenya National Bureau of Statistics (KNBS), vital registration, private health institutions, research institutions and FBOs amongst others.

RESPONSIBILITIES

1. NHIS Coordinating Committee (NHISCC)

- This is made up of representatives from senior level officers in the Ministry of Health, statistical constituencies and development partners.
- Concerned with provision of technical advisory role for health and social welfare data management in close collaboration with other strategic partners including KNBS and Vital Registration. The committee ensures unified and timely data collection, collation, processing and dissemination.

2. GENERAL PUBLIC

They ensure that any vital events or other significant health occurrences in the community are reported to the responsible authorities.

3. PRIVATE SECTOR

All health service providers in the private sector have a mandatory

responsibility to submit their health data regularly including data on all diseases under surveillance.

4. NON-GOVERNMENTAL ORGANISATIONS are responsible for ensuring that all health facilities under their respective umbrellas adhere to the NHIS. In collaboration with the Ministry of Health and other partners, NGOs are expected to mobilise resources for NHIS and ensure efficient data and information management in all its satellite facilities.

Ministry of Health - Roles by level

Community level

Every Community Unit (CU) should maintain and update its CHIS that shall be shared regularly with household members in a forum as stated in the health sector community strategy. The community health workers maintain registers recording daily activities and reporting regularly to supervising health facility.

HEALTH FACILITIES LEVEL

- maintain and update HIS data which include records, filing system(s) and registry for primary data collection tools such as registers, cards, file folders, summary forms such as reporting forms, CDs, electronic backups safeguarded from any risks e.g. fire, floods, access by unauthorised person, etc.
- Every health facility summarise health and health related data from the community and health facility, analyse, disseminate and use the information for decision-making, provide feedback then transmit summaries to the next level.

SUB-COUNTY LEVEL

has an oversight responsibility to manage all health and health related data from all service providers within their area of jurisdiction.

- i. It provides technical, material and financial support to all service providers in HIS.
- ii. Creates and maintains a data repository.
- iii. Collaborates and work in partnership with other statistical constituencies at the sub-county level to build one HIS.
- iv. It should collate, analyse, disseminate, use health and health related data from all health facilities/providers and give feedback to all health care providers.

COUNTY GOVERNMENT

has oversight responsibility to manage all health and health related data from all service providers within their area of jurisdiction.

- I. Provides technical, material and financial support to all sub-county government and service providers in HIS.
- II. Maintains a data repository for the county.
- III. Collaborates and works in partnership with other statistical constituencies at the provincial level to build one HIS.
- IV. Collates, analyses, disseminates and makes use health and health related data from all sub-counties and gives feedback to all health care providers in addition to submitting the same to MoH.

MOH

have oversight responsibility to manage all health and health related data from all service providers.

- I. Provides technical, material and financial support to all counties and service providers in HIS.
- II. Develops guidelines and formulates policies.
- III. Coordinates development of minimum data sets and data requirements of the health sector.
- IV. Collate, analyse, disseminate and use health and health related data from all counties and service providers and provide feedback to all.
- V. Creates and maintains a national data repository.
- VI. Collaborates and works in partnership with other statistical constituencies at the national to build one HIS.

BENEFITS OF HMIS

1. Helping decision makers to detect and control emerging and endemic health problems.
2. Help in monitoring progress towards health goals and promote equity.
3. Empowering individuals and communities with timely and understandable health related information.
4. Improving quality of services.
5. Strengthening the evidence base for effective health policies.
6. Permitting evaluation of scale up efforts and enabling innovation through research.
7. Mobilizing new resources and ensuring accountability in the way they are used.
8. Improving governance.

Challenges facing HIS in Kenya today

1. Competing priorities
2. Lack of or inadequacy of requisite skills at all levels
3. Inadequate personnel
4. Low staff morale, poor working and storage space at all level
5. Inadequate supply of equipments such computers and printers,
6. Challenges facing HIS in Kenya cont..
7. Lack of centralized data base at all levels
8. Weak feedback/data sharing mechanism and poor dissemination at all levels
9. Inadequate facilitative supervision
10. Inadequate finances/unclear funding mechanism
11. Lack of reporting tools
12. Late report and incomplete reports

Possible solutions of the challenges facing HIS in Kenya today

1. Develop staff skills and understanding
2. Intensify data sharing to create demand
3. Strengthen local information systems and structures
4. Create an information culture (Data Demand and Information Use)

THANK YOU